

---

## Density and Specific Gravity

---

### 2-1C

**Solution** We are to discuss the difference between intensive and extensive properties.

**Analysis** *Intensive properties do not depend on the size (extent) of the system but extensive properties do depend on the size (extent) of the system.*

**Discussion** An example of an intensive property is temperature. An example of an extensive property is mass.

---

### 2-2C

**Solution** We are to define specific gravity and discuss its relationship to density.

**Analysis** The *specific gravity*, or *relative density*, is defined as **the ratio of the density of a substance to the density of some standard substance at a specified temperature** (the standard is water at 4°C, for which  $\rho_{\text{H}_2\text{O}} = 1000 \text{ kg/m}^3$ ). That is,  $SG = \rho / \rho_{\text{H}_2\text{O}}$ . When specific gravity is known, density is determined from  $\rho = SG \times \rho_{\text{H}_2\text{O}}$ .

**Discussion** Specific gravity is dimensionless and unitless [it is just a number without dimensions or units].

---

### 2-3C

**Solution** We are to discuss the applicability of the ideal gas law.

**Analysis** A gas can be treated as an *ideal gas* when it is **at a high temperature and/or a low pressure relative to its critical temperature and pressure**.

**Discussion** Air and many other gases at room temperature and pressure can be approximated as ideal gases without any significant loss of accuracy.

---

### 2-4C

**Solution** We are to discuss the difference between  $R$  and  $R_u$ .

**Analysis**  $R_u$  is the **universal gas constant** that is the **same for all gases**, whereas  $R$  is the **specific gas constant** that is **different for different gases**. These two are related to each other by  $R = R_u / M$ , where  $M$  is the *molar mass* (also called the *molecular weight*) of the gas.

**Discussion** Since molar mass has dimensions of mass per mole,  $R$  and  $R_u$  do not have the same dimensions or units.

---

## 2-5

**Solution** A balloon is filled with helium gas. The number of moles and the mass of helium are to be determined.

**Assumptions** At specified conditions, helium behaves as an ideal gas.

**Properties** The universal gas constant is  $R_u = 8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K}$ . The molar mass of helium is  $4.0 \text{ kg/kmol}$ .

**Analysis** The volume of the sphere is

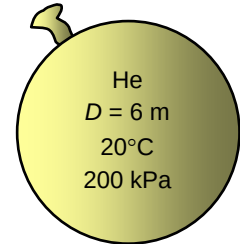
$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (3 \text{ m})^3 = 113.1 \text{ m}^3$$

Assuming ideal gas behavior, the number of moles of He is determined from

$$N = \frac{PV}{R_u T} = \frac{(200 \text{ kPa})(113.1 \text{ m}^3)}{(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(293 \text{ K})} = \mathbf{9.286 \text{ kmol}}$$

Then the mass of He is determined from

$$m = NM = (9.286 \text{ kmol})(4.0 \text{ kg/kmol}) = \mathbf{37.1 \text{ kg}}$$



**Discussion** Although the helium mass may seem large (about half the mass of an adult man!), it is much smaller than that of the air it displaces, and that is why helium balloons rise in the air.

## 2-6



**Solution** A balloon is filled with helium gas. The effect of the balloon diameter on the mass of helium is to be investigated, and the results are to be tabulated and plotted.

**Analysis** The EES *Equations* window is shown below, followed by the *Solution* window and the parametric table.

"Given Data"

{D=6"[m]"}

{P=200"[kPa]"}

T=20"[C]"

P=100"[kPa]"

R\_u=8.314"[kJ/kmol\*K]"

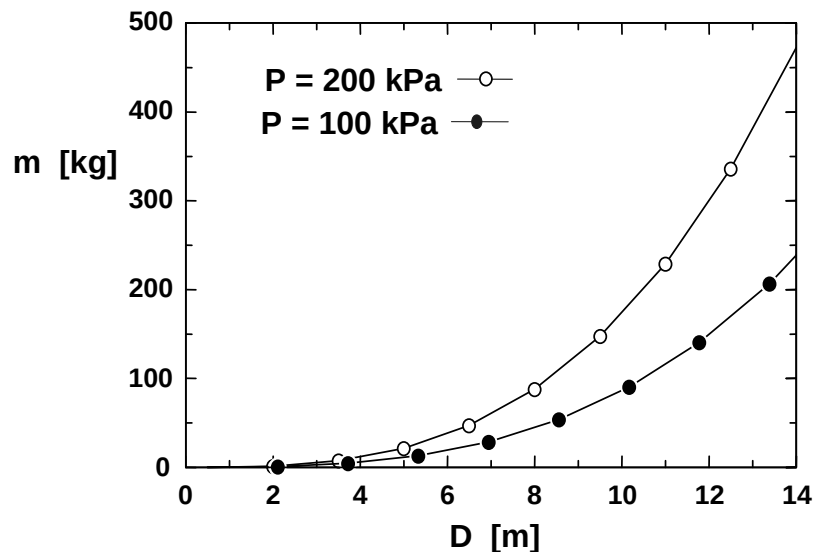
"Solution"

P\*V=N\*R\_u\*(T+273)

V=4\*pi\*(D/2)^3/3"[m^3]"

m=N\*MOLARMASS(Helium)"[kg]"

| D [m] | m [kg]  |
|-------|---------|
| 0.5   | 0.01075 |
| 2.111 | 0.8095  |
| 3.722 | 4.437   |
| 5.333 | 13.05   |
| 6.944 | 28.81   |
| 8.556 | 53.88   |
| 10.17 | 90.41   |
| 11.78 | 140.6   |
| 13.39 | 206.5   |
| 15    | 290.4   |



Mass of Helium in Balloon as function of Diameter

**Discussion** Mass increases with diameter as expected, but not linearly since volume is proportional to  $D^3$ .

## 2-7

**Solution** An automobile tire is inflated with air. The pressure rise of air in the tire when the tire is heated and the amount of air that must be bled off to reduce the temperature to the original value are to be determined.

**Assumptions** 1 At specified conditions, air behaves as an ideal gas. 2 The volume of the tire remains constant.

**Properties** The gas constant of air is  $R = 0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}$ .

**Analysis** Initially, the absolute pressure in the tire is

$$P_1 = P_g + P_{atm} = 210 + 100 = 310 \text{ kPa}$$

Treating air as an ideal gas and assuming the volume of the tire to remain constant, the final pressure in the tire is determined from

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \longrightarrow P_2 = \frac{T_2}{T_1} P_1 = \frac{323 \text{ K}}{298 \text{ K}} (310 \text{ kPa}) = 336 \text{ kPa}$$

Thus the pressure rise is

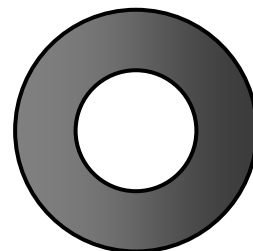
$$\Delta P = P_2 - P_1 = 336 - 310 = \mathbf{26.0 \text{ kPa}}$$

The amount of air that needs to be bled off to restore pressure to its original value is

$$m_1 = \frac{P_1 V}{RT_1} = \frac{(310 \text{ kPa})(0.025 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(298 \text{ K})} = 0.0906 \text{ kg}$$

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(336 \text{ kPa})(0.025 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(323 \text{ K})} = 0.0836 \text{ kg}$$

$$\Delta m = m_1 - m_2 = 0.0906 - 0.0836 = \mathbf{0.0070 \text{ kg}}$$



Tire  
25°C  
210 kPa

**Discussion** Notice that *absolute* rather than gage pressure must be used in calculations with the ideal gas law.

## 2-8E

**Solution** An automobile tire is under-inflated with air. The amount of air that needs to be added to the tire to raise its pressure to the recommended value is to be determined.

**Assumptions** 1 At specified conditions, air behaves as an ideal gas. 2 The volume of the tire remains constant.

**Properties** The gas constant of air is  $R = 0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R}$ .

**Analysis** The initial and final absolute pressures in the tire are

$$P_1 = P_{g1} + P_{atm} = 20 + 14.6 = 34.6 \text{ psia}$$

$$P_2 = P_{g2} + P_{atm} = 30 + 14.6 = 44.6 \text{ psia}$$

Treating air as an ideal gas, the initial mass in the tire is

$$m_1 = \frac{P_1 V}{RT_1} = \frac{(34.6 \text{ psia})(0.53 \text{ ft}^3)}{(0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R})(550 \text{ R})} = 0.0900 \text{ lbm}$$

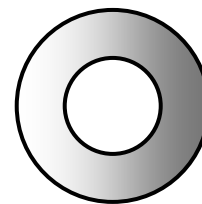
Noting that the temperature and the volume of the tire remain constant, the final mass in the tire becomes

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(44.6 \text{ psia})(0.53 \text{ ft}^3)}{(0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R})(550 \text{ R})} = 0.1160 \text{ lbm}$$

Thus the amount of air that needs to be added is  $\Delta m = m_2 - m_1 = 0.1160 - 0.0900 = \mathbf{0.0260 \text{ lbm}}$

**Discussion** Notice that *absolute* rather than gage pressure must be used in calculations with the ideal gas law.

Tire  
0.53 ft<sup>3</sup>  
90°F  
20 psia



**2-9E**

**Solution** A rigid tank contains slightly pressurized air. The amount of air that needs to be added to the tank to raise its pressure and temperature to the recommended values is to be determined.

**Assumptions** **1** At specified conditions, air behaves as an ideal gas. **2** The volume of the tank remains constant.

**Properties** The gas constant of air is  $R = 0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R}$ .

**Analysis** Treating air as an ideal gas, the initial volume and the final mass in the tank are determined to be

$$V = \frac{m_1 R T_1}{P_1} = \frac{(20 \text{ lbm})(0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R})(530 \text{ R})}{20 \text{ psia}} = 196.3 \text{ ft}^3$$

$$m_2 = \frac{P_2 V}{R T_2} = \frac{(35 \text{ psia})(196.3 \text{ ft}^3)}{(0.3704 \text{ psia} \cdot \text{ft}^3/\text{lbm} \cdot \text{R})(550 \text{ R})} = 33.73 \text{ lbm}$$

Thus the amount of air added is

$$\Delta m = m_2 - m_1 = 33.73 - 20.0 = \mathbf{13.7 \text{ lbm}}$$

|                                |
|--------------------------------|
| Air, 20 lbm<br>20 psia<br>70°F |
|--------------------------------|

**Discussion** As the temperature slowly decreases due to heat transfer, the pressure will also decrease.

---

## 2-10

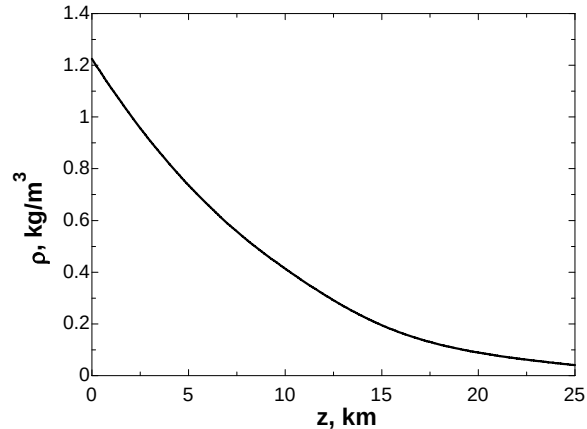


**Solution** A relation for the variation of density with elevation is to be obtained, the density at 7 km elevation is to be calculated, and the mass of the atmosphere using the correlation is to be estimated.

**Assumptions** 1 Atmospheric air behaves as an ideal gas. 2 The earth is perfectly spherical with a radius of 6377 km at sea level, and the thickness of the atmosphere is 25 km.

**Properties** The density data are given in tabular form as a function of radius and elevation, where  $r = z + 6377$  km:

| $r$ , km | $z$ , km | $\rho$ , kg/m <sup>3</sup> |
|----------|----------|----------------------------|
| 6377     | 0        | 1.225                      |
| 6378     | 1        | 1.112                      |
| 6379     | 2        | 1.007                      |
| 6380     | 3        | 0.9093                     |
| 6381     | 4        | 0.8194                     |
| 6382     | 5        | 0.7364                     |
| 6383     | 6        | 0.6601                     |
| 6385     | 8        | 0.5258                     |
| 6387     | 10       | 0.4135                     |
| 6392     | 15       | 0.1948                     |
| 6397     | 20       | 0.08891                    |
| 6402     | 25       | 0.04008                    |



**Analysis** Using EES, (1) Define a trivial function “rho= a+ z” in the Equation window, (2) select new parametric table from Tables, and type the data in a two-column table, (3) select Plot and plot the data, and (4) select Plot and click on curve fit to get curve fit window. Then specify 2<sup>nd</sup> order polynomial and enter/edit equation. The results are:

$$\rho(z) = a + bz + cz^2 = 1.20252 - 0.101674z + 0.0022375z^2 \text{ for the unit of kg/m}^3,$$

$$(\text{or, } \rho(z) = (1.20252 - 0.101674z + 0.0022375z^2) \times 10^9 \text{ for the unit of kg/km}^3)$$

where  $z$  is the vertical distance from the earth surface at sea level. At  $z = 7$  km, the equation gives  $\rho = 0.600 \text{ kg/m}^3$ .

(b) The mass of atmosphere is evaluated by integration to be

$$m = \int_V \rho dV = \int_{z=0}^h (a + bz + cz^2) 4\pi(r_0 + z)^2 dz = 4\pi \int_{z=0}^h (a + bz + cz^2)(r_0^2 + 2r_0z + z^2) dz$$

$$= 4\pi \left[ ar_0^2 h + r_0(2a + br_0)h^2 / 2 + (a + 2br_0 + cr_0^2)h^3 / 3 + (b + 2cr_0)h^4 / 4 + ch^5 / 5 \right]$$

where  $r_0 = 6377$  km is the radius of the earth,  $h = 25$  km is the thickness of the atmosphere. Also,  $a = 1.20252$ ,  $b = -0.101674$ , and  $c = 0.0022375$  are the constants in the density function. Substituting and multiplying by the factor  $10^9$  to convert the density from units of  $\text{kg/km}^3$  to  $\text{kg/m}^3$ , the mass of the atmosphere is determined to be approximately

$$m = 5.09 \times 10^{18} \text{ kg}$$

EES Solution for final result:

$$a=1.2025166$$

$$b=-0.10167$$

$$c=0.0022375$$

$$r=6377$$

$$h=25$$

$$m=4*\pi*(a*r^2*h+r*(2*a+b*r)*h^2/2+(a+2*b*r+c*r^2)*h^3/3+(b+2*c*r)*h^4/4+c*h^5/5)*1E+9$$

**Discussion** At 7 km, the density of the air is approximately half of its value at sea level.

---

## Vapor Pressure and Cavitation

---

### 2-11C

**Solution** We are to define vapor pressure and discuss its relationship to saturation pressure.

**Analysis** The *vapor pressure*  $P_v$  of a pure substance is defined as **the pressure exerted by a vapor in phase equilibrium with its liquid at a given temperature**. In general, the pressure of a vapor or gas, whether it exists alone or in a mixture with other gases, is called the *partial pressure*. During phase change processes between the liquid and vapor phases of a pure substance, the **saturation pressure and the vapor pressure are equivalent** since the vapor is pure.

**Discussion** Partial pressure is not necessarily equal to vapor pressure. For example, on a dry day (low relative humidity), the partial pressure of water vapor in the air is less than the vapor pressure of water. If, however, the relative humidity is 100%, the partial pressure and the vapor pressure are equal.

---

### 2-12C

**Solution** We are to discuss whether the boiling temperature of water increases as pressure increases.

**Analysis** **Yes**. The saturation temperature of a pure substance depends on pressure; in fact, it *increases* with pressure. The higher the pressure, the higher the saturation or boiling temperature.

**Discussion** This fact is easily seen by looking at the saturated water property tables. Note that boiling temperature and saturation pressure at a given pressure are equivalent.

---

### 2-13C

**Solution** We are to determine if temperature increases or remains constant when the pressure of a boiling substance increases.

**Analysis** If the pressure of a substance increases during a boiling process, **the temperature also increases** since the boiling (or saturation) temperature of a pure substance depends on pressure and increases with it.

**Discussion** We are assuming that the liquid will continue to boil. If the pressure is increased fast enough, boiling may stop until the temperature has time to reach its new (higher) boiling temperature. A pressure cooker uses this principle.

---

### 2-14C

**Solution** We are to define and discuss cavitation.

**Analysis** In the flow of a liquid, *cavitation* is the **vaporization that may occur at locations where the pressure drops below the vapor pressure**. The vapor bubbles collapse as they are swept away from the low pressure regions, generating highly destructive, extremely high-pressure waves. This phenomenon is a common cause for drop in performance and even the erosion of impeller blades.

**Discussion** The word “cavitation” comes from the fact that a vapor bubble or “cavity” appears in the liquid. Not all cavitation is undesirable. It turns out that some underwater vehicles employ “super cavitation” on purpose to reduce drag.

---

## 2-15

**Solution** The minimum pressure in a piping system to avoid cavitation is to be determined.

**Properties** The vapor pressure of water at 40°C is 7.38 kPa.

**Analysis** To avoid cavitation, the pressure anywhere in the flow should not be allowed to drop below the vapor (or saturation) pressure at the given temperature. That is,

$$P_{\min} = P_{\text{sat}@40^\circ\text{C}} = \mathbf{7.38 \text{ kPa}}$$

Therefore, the pressure should be maintained above 7.38 kPa everywhere in flow.

**Discussion** Note that the vapor pressure increases with increasing temperature, and thus the risk of cavitation is greater at higher fluid temperatures.

---

## 2-16

**Solution** The minimum pressure in a pump is given. It is to be determined if there is a danger of cavitation.

**Properties** The vapor pressure of water at 20°C is 2.339 kPa.

**Analysis** To avoid cavitation, the pressure everywhere in the flow should remain above the vapor (or saturation) pressure at the given temperature, which is

$$P_v = P_{\text{sat}@20^\circ\text{C}} = 2.339 \text{ kPa}$$

The minimum pressure in the pump is 2 kPa, which is less than the vapor pressure. Therefore, a **there is danger of cavitation in the pump**.

**Discussion** Note that the vapor pressure increases with increasing temperature, and thus there is a greater danger of cavitation at higher fluid temperatures.

---

## 2-17E

**Solution** The minimum pressure in a pump is given. It is to be determined if there is a danger of cavitation.

**Properties** The vapor pressure of water at 70°F is 0.3632 psia.

**Analysis** To avoid cavitation, the pressure everywhere in the flow should remain above the vapor (or saturation) pressure at the given temperature, which is

$$P_v = P_{\text{sat}@70^\circ\text{F}} = 0.3632 \text{ psia}$$

The minimum pressure in the pump is 0.1 psia, which is less than the vapor pressure. Therefore, **there is danger of cavitation in the pump**.

**Discussion** Note that the vapor pressure increases with increasing temperature, and the danger of cavitation increases at higher fluid temperatures.

---

## 2-18

**Solution** The minimum pressure in a pump to avoid cavitation is to be determined.

**Properties** The vapor pressure of water at 25°C is 3.17 kPa.

**Analysis** To avoid cavitation, the pressure anywhere in the system should not be allowed to drop below the vapor (or saturation) pressure at the given temperature. That is,

$$P_{\min} = P_{\text{sat}@25^\circ\text{C}} = \mathbf{3.17 \text{ kPa}}$$

Therefore, the lowest pressure that can exist in the pump is 3.17 kPa.

**Discussion** Note that the vapor pressure increases with increasing temperature, and thus the risk of cavitation is greater at higher fluid temperatures.

---

---

## Energy and Specific Heats

---

### 2-19C

**Solution** We are to discuss the difference between macroscopic and microscopic forms of energy.

**Analysis** The *macroscopic* forms of energy are those **a system possesses as a whole** with respect to some outside reference frame. The *microscopic* forms of energy, on the other hand, are those **related to the molecular structure of a system and the degree of the molecular activity**, and are independent of outside reference frames.

**Discussion** We mostly deal with macroscopic forms of energy in fluid mechanics.

---

### 2-20C

**Solution** We are to define total energy and identify its constituents.

**Analysis** The **sum of all forms of the energy a system possesses** is called *total energy*. In the absence of magnetic, electrical, and surface tension effects, **the total energy of a system consists of the kinetic, potential, and internal energies**.

**Discussion** All three constituents of total energy (kinetic, potential, and internal) need to be considered in an analysis of a general fluid flow.

---

### 2-21C

**Solution** We are to list the forms of energy that contribute to the internal energy of a system.

**Analysis** The *internal energy* of a system is made up of **sensible, latent, chemical, and nuclear energies**. The sensible internal energy is due to translational, rotational, and vibrational effects.

**Discussion** We deal with the flow of a single phase fluid in most problems in this textbook; therefore, latent, chemical, and nuclear energies do not need to be considered.

---

### 2-22C

**Solution** We are to discuss the relationship between heat, internal energy, and thermal energy.

**Analysis** *Thermal energy* is the **sensible and latent forms of internal energy**. It does not include chemical or nuclear forms of energy. In common terminology, thermal energy is referred to as *heat*. However, like work, heat is not a property, whereas thermal energy is a property.

**Discussion** Technically speaking, “heat” is defined only when there is *heat transfer*, whereas the energy state of a substance can always be defined, even if no heat transfer is taking place.

---

### 2-23C

**Solution** We are to define and discuss flow energy.

**Analysis** *Flow energy* or *flow work* is the **energy needed to push a fluid into or out of a control volume**. Fluids at rest do not possess any flow energy.

**Discussion** Flow energy is not a fundamental quantity, like kinetic or potential energy. However, it is a useful concept in fluid mechanics since fluids are often forced into and out of control volumes in practice.

---



## 2-24C

**Solution** We are to compare the energies of flowing and non-flowing fluids.

**Analysis** A flowing fluid possesses **flow energy, which is the energy needed to push a fluid into or out of a control volume**, in addition to the forms of energy possessed by a non-flowing fluid. The total energy of a non-flowing fluid consists of internal and potential energies. If the fluid is moving as a rigid body, but not *flowing*, it may also have kinetic energy (e.g., gasoline in a tank truck moving down the highway at constant speed with no sloshing). The total energy of a flowing fluid consists of internal, kinetic, potential, *and* flow energies.

**Discussion** Flow energy is not to be confused with kinetic energy, even though both are zero when the fluid is at rest.

---

## 2-25C

**Solution** We are to explain how changes in internal energy can be determined.

**Analysis** Using specific heat values at the average temperature, the changes in the specific internal energy of ideal gases can be determined from  $\Delta u = c_{v,avg} \Delta T$ . For incompressible substances,  $c_p \cong c_v \cong c$  and  $\Delta u = c_{avg} \Delta T$ .

**Discussion** If the fluid can be treated as neither incompressible nor an ideal gas, property tables must be used.

---

## 2-26C

**Solution** We are to explain how changes in enthalpy can be determined.

**Analysis** Using specific heat values at the average temperature, the changes in specific enthalpy of ideal gases can be determined from  $\Delta h = c_{p,avg} \Delta T$ . For incompressible substances,  $c_p \cong c_v \cong c$  and  $\Delta h = \Delta u + v\Delta P \cong c_{avg} \Delta T + v\Delta P$ .

**Discussion** If the fluid can be treated as neither incompressible nor an ideal gas, property tables must be used.

---

---

## Coefficient of Compressibility

---

## 2-27C

**Solution** We are to discuss the coefficient of compressibility and the isothermal compressibility.

**Analysis** The *coefficient of compressibility* represents the **variation of pressure of a fluid with volume or density at constant temperature**. *Isothermal compressibility* is the inverse of the coefficient of compressibility, and it represents the **fractional change in volume or density corresponding to a change in pressure**.

**Discussion** The coefficient of compressibility of an ideal gas is equal to its absolute pressure.

---

## 2-28C

**Solution** We are to define the coefficient of volume expansion.

**Analysis** The *coefficient of volume expansion* represents the **variation of the density of a fluid with temperature at constant pressure**. It differs from the coefficient of compressibility in that the latter represents the variation of *pressure* of a fluid with density at constant temperature.

**Discussion** The coefficient of volume expansion of an ideal gas is equal to the inverse of its absolute temperature.

---

## 2-29C

**Solution** We are to discuss the sign of the coefficient of compressibility and the coefficient of volume expansion.

**Analysis** The *coefficient of compressibility* of a fluid **cannot be negative**, but the *coefficient of volume expansion* **can be negative** (e.g., liquid water below 4°C).

**Discussion** This is the reason that ice floats on water.

---

## 2-30

**Solution** The percent increase in the density of an ideal gas is given for a moderate pressure. The percent increase in density of the gas when compressed at a higher pressure is to be determined.

**Assumptions** The gas behaves an ideal gas.

**Analysis** For an ideal gas,  $P = \rho RT$  and  $(\partial P / \partial \rho)_T = RT = P / \rho$ , and thus  $\kappa_{\text{ideal gas}} = P$ . Therefore, the coefficient of compressibility of an ideal gas is equal to its absolute pressure, and the coefficient of compressibility of the gas increases with increasing pressure.

Substituting  $\kappa = P$  into the definition of the coefficient of compressibility  $\kappa \cong -\frac{\Delta P}{\Delta v / v} \cong \frac{\Delta P}{\Delta \rho / \rho}$  and rearranging gives

$$\frac{\Delta \rho}{\rho} = \frac{\Delta P}{P}$$

Therefore, **the percent increase of density of an ideal gas during isothermal compression is equal to the percent increase in pressure.**

$$\text{At 10 atm: } \frac{\Delta \rho}{\rho} = \frac{\Delta P}{P} = \frac{11-10}{10} = 10\%$$

$$\text{At 100 atm: } \frac{\Delta \rho}{\rho} = \frac{\Delta P}{P} = \frac{101-100}{100} = 1\%$$

Therefore, a pressure change of 1 atm causes a density change of 10% at 10 atm and a density change of 1% at 100 atm.

**Discussion** If temperature were also allowed to change, the relationship would not be so simple.

---

## 2-31

**Solution** Using the definition of the coefficient of volume expansion and the expression  $\beta_{\text{ideal gas}} = 1/T$ , it is to be shown that the percent increase in the specific volume of an ideal gas during isobaric expansion is equal to the percent increase in absolute temperature.

**Assumptions** The gas behaves an ideal gas.

**Analysis** The coefficient of volume expansion  $\beta$  can be expressed as  $\beta = \frac{1}{v} \left( \frac{\partial v}{\partial T} \right)_P \approx \frac{\Delta v / v}{\Delta T}$ .

Noting that  $\beta_{\text{ideal gas}} = 1/T$  for an ideal gas and rearranging give

$$\frac{\Delta v}{v} = \frac{\Delta T}{T}$$

Therefore, **the percent increase in the specific volume of an ideal gas during isobaric expansion is equal to the percent increase in absolute temperature.**

**Discussion** We must be careful to use *absolute* temperature (K or R), not relative temperature (°C or °F).

---

## 2-32

**Solution** Water at a given temperature and pressure is compressed to a high pressure isothermally. The increase in the density of water is to be determined.

**Assumptions** 1 The isothermal compressibility is constant in the given pressure range. 2 An approximate analysis is performed by replacing differential changes by finite changes.

**Properties** The density of water at 20°C and 1 atm pressure is  $\rho_1 = 998 \text{ kg/m}^3$ . The isothermal compressibility of water is given to be  $\alpha = 4.80 \times 10^{-5} \text{ atm}^{-1}$ .

**Analysis** When differential quantities are replaced by differences and the properties  $\alpha$  and  $\beta$  are assumed to be constant, the change in density in terms of the changes in pressure and temperature is expressed approximately as

$$\Delta\rho = \alpha\rho\Delta P - \beta\rho\Delta T$$

The change in density due to a change of pressure from 1 atm to 800 atm at constant temperature is

$$\Delta\rho = \alpha\rho\Delta P = (4.80 \times 10^{-5} \text{ atm}^{-1})(998 \text{ kg/m}^3)(800 - 1)\text{atm} = \mathbf{38.3 \text{ kg/m}^3}$$

**Discussion** Note that the density of water increases from 998 to 1036.3 kg/m<sup>3</sup> while being compressed, as expected. This problem can be solved more accurately using differential analysis when functional forms of properties are available.

---

## 2-33

**Solution** Water at a given temperature and pressure is heated to a higher temperature at constant pressure. The change in the density of water is to be determined.

**Assumptions** 1 The coefficient of volume expansion is constant in the given temperature range. 2 An approximate analysis is performed by replacing differential changes in quantities by finite changes.

**Properties** The density of water at 15°C and 1 atm pressure is  $\rho_1 = 999.1 \text{ kg/m}^3$ . The coefficient of volume expansion at the average temperature of  $(15+95)/2 = 55^\circ\text{C}$  is  $\beta = 0.484 \times 10^{-3} \text{ K}^{-1}$ .

**Analysis** When differential quantities are replaced by differences and the properties  $\alpha$  and  $\beta$  are assumed to be constant, the change in density in terms of the changes in pressure and temperature is expressed approximately as

$$\Delta\rho = \alpha\rho\Delta P - \beta\rho\Delta T$$

The change in density due to the change of temperature from 15°C to 95°C at constant pressure is

$$\Delta\rho = -\beta\rho\Delta T = -(0.484 \times 10^{-3} \text{ K}^{-1})(999.1 \text{ kg/m}^3)(95 - 15)\text{K} = \mathbf{-38.7 \text{ kg/m}^3}$$

**Discussion** Noting that  $\Delta\rho = \rho_2 - \rho_1$ , the density of water at 95°C and 1 atm is

$$\rho_2 = \rho_1 + \Delta\rho = 999.1 + (-38.7) = 960.4 \text{ kg/m}^3$$

which is very close to the listed value of 961.5 kg/m<sup>3</sup> at 95°C in water table in the Appendix. This is mostly due to  $\beta$  varying with temperature almost linearly. Note that the density of water decreases while being heated, as expected. This problem can be solved more accurately using differential analysis when functional forms of properties are available.

---

**2-34**

**Solution** Saturated refrigerant-134a at a given temperature is cooled at constant pressure. The change in the density of the refrigerant is to be determined.

**Assumptions** 1 The coefficient of volume expansion is constant in the given temperature range. 2 An approximate analysis is performed by replacing differential changes in quantities by finite changes.

**Properties** The density of saturated liquid R-134a at 10°C is  $\rho_1 = 1261 \text{ kg/m}^3$ . The coefficient of volume expansion at the average temperature of  $(10+0)/2 = 5^\circ\text{C}$  is  $\beta = 0.00269 \text{ K}^{-1}$ .

**Analysis** When differential quantities are replaced by differences and the properties  $\alpha$  and  $\beta$  are assumed to be constant, the change in density in terms of the changes in pressure and temperature is expressed approximately as

$$\Delta\rho = \alpha\rho\Delta P - \beta\rho\Delta T$$

The change in density due to the change of temperature from 10°C to 0°C at constant pressure is

$$\Delta\rho = -\beta\rho\Delta T = -(0.00269 \text{ K}^{-1})(1261 \text{ kg/m}^3)(0-10)\text{K} = \mathbf{33.9 \text{ kg/m}^3}$$

**Discussion** Noting that  $\Delta\rho = \rho_2 - \rho_1$ , the density of R-134a at 0°C is

$$\rho_2 = \rho_1 + \Delta\rho = 1261 + 33.9 = 1294.9 \text{ kg/m}^3$$

which is almost identical to the listed value of 1295 kg/m<sup>3</sup> at 0°C in R-134a table in the Appendix. This is mostly due to  $\beta$  varying with temperature almost linearly. Note that the density increases during cooling, as expected.

---

**2-35**

**Solution** A water tank completely filled with water can withstand tension caused by a volume expansion of 2%. The maximum temperature rise allowed in the tank without jeopardizing safety is to be determined.

**Assumptions** 1 The coefficient of volume expansion is constant. 2 An approximate analysis is performed by replacing differential changes in quantities by finite changes. 3 The effect of pressure is disregarded.

**Properties** The average volume expansion coefficient is given to be  $\beta = 0.377 \times 10^{-3} \text{ K}^{-1}$ .

**Analysis** When differential quantities are replaced by differences and the properties  $\alpha$  and  $\beta$  are assumed to be constant, the change in density in terms of the changes in pressure and temperature is expressed approximately as

$$\Delta\rho = \alpha\rho\Delta P - \beta\rho\Delta T$$

A volume increase of 2% corresponds to a density decrease of 2%, which can be expressed as  $\Delta\rho = -0.02\rho$ . Then the decrease in density due to a temperature rise of  $\Delta T$  at constant pressure is

$$-0.02\rho = -\beta\rho\Delta T$$

Solving for  $\Delta T$  and substituting, the maximum temperature rise is determined to be

$$\Delta T = \frac{0.02}{\beta} = \frac{0.02}{0.377 \times 10^{-3} \text{ K}^{-1}} = \mathbf{53.0 \text{ K} = 53.0^\circ\text{C}}$$

**Discussion** This result is conservative since in reality the increasing pressure will tend to compress the water and increase its density.

---

**2-36**

**Solution** A water tank completely filled with water can withstand tension caused by a volume expansion of 1%. The maximum temperature rise allowed in the tank without jeopardizing safety is to be determined.

**Assumptions** 1 The coefficient of volume expansion is constant. 2 An approximate analysis is performed by replacing differential changes in quantities by finite changes. 3 The effect of pressure is disregarded.

**Properties** The average volume expansion coefficient is given to be  $\beta = 0.377 \times 10^{-3} \text{ K}^{-1}$ .

**Analysis** When differential quantities are replaced by differences and the properties  $\alpha$  and  $\beta$  are assumed to be constant, the change in density in terms of the changes in pressure and temperature is expressed approximately as

$$\Delta\rho = \alpha\rho\Delta P - \beta\rho\Delta T$$

A volume increase of 1% corresponds to a density decrease of 1%, which can be expressed as  $\Delta\rho = -0.01\rho$ . Then the decrease in density due to a temperature rise of  $\Delta T$  at constant pressure is

$$-0.01\rho = -\beta\rho\Delta T$$

Solving for  $\Delta T$  and substituting, the maximum temperature rise is determined to be

$$\Delta T = \frac{0.01}{\beta} = \frac{0.01}{0.377 \times 10^{-3} \text{ K}^{-1}} = \mathbf{26.5 \text{ K} = 26.5^\circ\text{C}}$$

**Discussion** This result is conservative since in reality the increasing pressure will tend to compress the water and increase its density. The change in temperature is exactly half of that of the previous problem, as expected.

---

## 2-37

**Solution** The density of seawater at the free surface and the bulk modulus of elasticity are given. The density and pressure at a depth of 2500 m are to be determined.

**Assumptions** 1 The temperature and the bulk modulus of elasticity of seawater is constant. 2 The gravitational acceleration remains constant.

**Properties** The density of seawater at free surface where the pressure is given to be  $1030 \text{ kg/m}^3$ , and the bulk modulus of elasticity of seawater is given to be  $2.34 \times 10^9 \text{ N/m}^2$ .

**Analysis** The coefficient of compressibility or the bulk modulus of elasticity of fluids is expressed as

$$\kappa = \rho \left( \frac{\partial P}{\partial \rho} \right)_T \quad \text{or} \quad \kappa = \rho \frac{dP}{d\rho} \quad (\text{at constant } T)$$

The differential pressure change across a differential fluid height of  $dz$  is given as

$$dP = \rho g dz$$

Combining the two relations above and rearranging,

$$\kappa = \rho \frac{\rho g dz}{d\rho} = g \rho^2 \frac{dz}{d\rho} \quad \rightarrow \quad \frac{d\rho}{\rho^2} = \frac{g dz}{\kappa}$$

Integrating from  $z = 0$  where  $\rho = \rho_0 = 1030 \text{ kg/m}^3$  to  $z = z$  where  $\rho = \rho$  gives

$$\int_{\rho_0}^{\rho} \frac{d\rho}{\rho^2} = \frac{g}{\kappa} \int_0^z dz \quad \rightarrow \quad \frac{1}{\rho_0} - \frac{1}{\rho} = \frac{gz}{\kappa}$$

Solving for  $\rho$  gives the variation of density with depth as

$$\rho = \frac{1}{(1/\rho_0) - (gz/\kappa)}$$

Substituting into the pressure change relation  $dP = \rho g dz$  and integrating from  $z = 0$  where  $P = P_0 = 98 \text{ kPa}$  to  $z = z$  where  $P = P$  gives

$$\int_{P_0}^P dP = \int_0^z \frac{g dz}{(1/\rho_0) - (gz/\kappa)} \quad \rightarrow \quad P = P_0 + \kappa \ln \left( \frac{1}{1 - (\rho_0 g z / \kappa)} \right)$$

which is the desired relation for the variation of pressure in seawater with depth. At  $z = 2500 \text{ m}$ , the values of density and pressure are determined by substitution to be

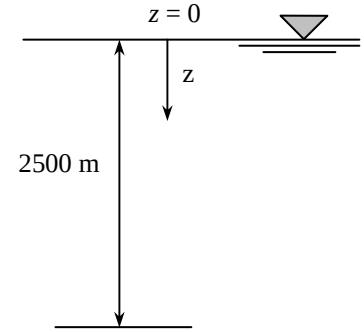
$$\rho = \frac{1}{1/(1030 \text{ kg/m}^3) - (9.81 \text{ m/s}^2)(2500 \text{ m})/(2.34 \times 10^9 \text{ N/m}^2)} = \mathbf{1041 \text{ kg/m}^3}$$

$$\begin{aligned} P &= (98,000 \text{ Pa}) + (2.34 \times 10^9 \text{ N/m}^2) \ln \left( \frac{1}{1 - (1030 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(2500 \text{ m})/(2.34 \times 10^9 \text{ N/m}^2)} \right) \\ &= 2.550 \times 10^7 \text{ Pa} \\ &= \mathbf{25.50 \text{ MPa}} \end{aligned}$$

since  $1 \text{ Pa} = 1 \text{ N/m}^2 = 1 \text{ kg/m} \cdot \text{s}^{-2}$  and  $1 \text{ kPa} = 1000 \text{ Pa}$ .

**Discussion** Note that if we assumed  $\rho = \rho_0 = \text{constant}$  at  $1030 \text{ kg/m}^3$ , the pressure at 2500 m would be  $P = P_0 + \rho g z = 0.098 + 25.26 = 25.36 \text{ MPa}$ . Then the density at 2500 m is estimated to be

$$\Delta \rho = \rho \alpha \Delta P = (1030)(2340 \text{ MPa})^{-1}(25.26 \text{ MPa}) = 11.1 \text{ kg/m}^3 \text{ and thus } \rho = 1041 \text{ kg/m}^3$$



---

## Viscosity

---

### 2-38C

**Solution** We are to define and discuss viscosity.

**Analysis** Viscosity is a measure of the “stickiness” or “resistance to deformation” of a fluid. It is due to the internal frictional force that develops between different layers of fluids as they are forced to move relative to each other. Viscosity is caused by the cohesive forces between the molecules in liquids, and by the molecular collisions in gases. In general, **liquids have higher dynamic viscosities than gases.**

**Discussion** The ratio of viscosity  $\mu$  to density  $\rho$  often appears in the equations of fluid mechanics, and is defined as the *kinematic viscosity*,  $\nu = \mu / \rho$ .

---

### 2-39C

**Solution** We are to discuss Newtonian fluids.

**Analysis** **Fluids whose shear stress is linearly proportional to the velocity gradient** (shear strain) are called *Newtonian fluids*. Most common fluids such as water, air, gasoline, and oils are Newtonian fluids.

**Discussion** In the differential analysis of fluid flow, only Newtonian fluids are considered in this textbook.

---

### 2-40C

**Solution** We are to compare the settling speed of balls dropped in water and oil; namely, we are to determine which will reach the bottom of the container first.

**Analysis** When two identical small glass balls are dropped into two identical containers, one filled with water and the other with oil, **the ball dropped in water will reach the bottom of the container first** because of the much lower viscosity of water relative to oil.

**Discussion** Oil is very viscous, with typical values of viscosity approximately 800 times greater than that of water at room temperature.

---

### 2-41C

**Solution** We are to discuss how dynamic viscosity varies with temperature in liquids and gases.

**Analysis** (a) The **dynamic viscosity of liquids decreases with temperature.** (b) The **dynamic viscosity of gases increases with temperature.**

**Discussion** A good way to remember this is that a car engine is much harder to start in the winter because the oil in the engine has a higher viscosity at low temperatures.

---

### 2-42C

**Solution** We are to discuss how kinematic viscosity varies with temperature in liquids and gases.

**Analysis** (a) **For liquids, the kinematic viscosity decreases with temperature.** (b) **For gases, the kinematic viscosity increases with temperature.**

**Discussion** You can easily verify this by looking at the appendices.

---

## 2-43

**Solution** A block is moved at constant velocity on an inclined surface. The force that needs to be applied in the horizontal direction when the block is dry, and the percent reduction in the required force when an oil film is applied on the surface are to be determined.

**Assumptions** 1 The inclined surface is plane (perfectly flat, although tilted). 2 The friction coefficient and the oil film thickness are uniform. 3 The weight of the oil layer is negligible.

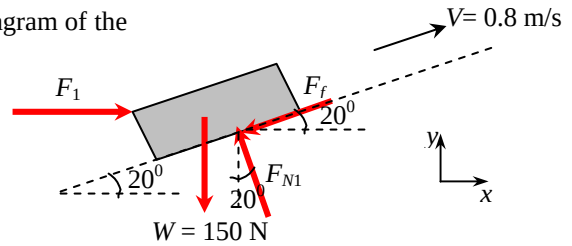
**Properties** The absolute viscosity of oil is given to be  $\mu = 0.012 \text{ Pa}\cdot\text{s} = 0.012 \text{ N}\cdot\text{s}/\text{m}^2$ .

**Analysis** (a) The velocity of the block is constant, and thus its acceleration and the net force acting on it are zero. A free body diagram of the block is given. Then the force balance gives

$$\sum F_x = 0: F_1 - F_f \cos 20^\circ - F_{N1} \sin 20^\circ = 0 \quad (1)$$

$$\sum F_y = 0: F_{N1} \cos 20^\circ - F_f \sin 20^\circ - W = 0 \quad (2)$$

$$\text{Friction force: } F_f = fF_{N1} \quad (3)$$



Substituting Eq. (3) into Eq. (2) and solving for  $F_{N1}$  gives

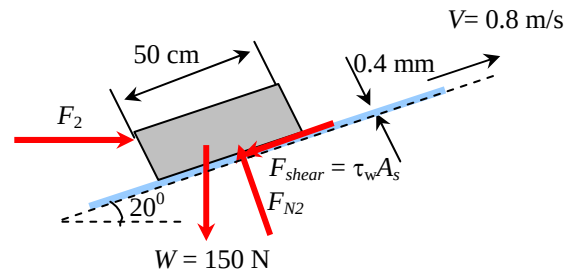
$$F_{N1} = \frac{W}{\cos 20^\circ - f \sin 20^\circ} = \frac{150 \text{ N}}{\cos 20^\circ - 0.27 \sin 20^\circ} = 177.0 \text{ N}$$

Then from Eq. (1):

$$F_1 = F_f \cos 20^\circ + F_{N1} \sin 20^\circ = (0.27 \times 177 \text{ N}) \cos 20^\circ + (177 \text{ N}) \sin 20^\circ = \mathbf{105.5 \text{ N}}$$

(b) In this case, the friction force is replaced by the shear force applied on the bottom surface of the block due to the oil. Because of the no-slip condition, the oil film sticks to the inclined surface at the bottom and the lower surface of the block at the top. Then the shear force is expressed as

$$\begin{aligned} F_{\text{shear}} &= \tau_w A_s \\ &= \mu A_s \frac{V}{h} \\ &= (0.012 \text{ N}\cdot\text{s}/\text{m}^2) (0.5 \times 0.2 \text{ m}^2) \frac{0.8 \text{ m/s}}{4 \times 10^{-4} \text{ m}} \\ &= 2.4 \text{ N} \end{aligned}$$



Replacing the friction force by the shear force in part (a),

$$\sum F_x = 0: F_2 - F_{\text{shear}} \cos 20^\circ - F_{N2} \sin 20^\circ = 0 \quad (4)$$

$$\sum F_y = 0: F_{N2} \cos 20^\circ - F_{\text{shear}} \sin 20^\circ - W = 0 \quad (5)$$

Eq. (5) gives  $F_{N2} = (F_{\text{shear}} \sin 20^\circ + W) / \cos 20^\circ = [(2.4 \text{ N}) \sin 20^\circ + (150 \text{ N})] / \cos 20^\circ = 160.5 \text{ N}$

Substituting into Eq. (4), the required horizontal force is determined to be

$$F_2 = F_{\text{shear}} \cos 20^\circ + F_{N2} \sin 20^\circ = (2.4 \text{ N}) \cos 20^\circ + (160.5 \text{ N}) \sin 20^\circ = 57.2 \text{ N}$$

Then, our final result is expressed as

$$\text{Percentage reduction in required force} = \frac{F_1 - F_2}{F_1} \times 100\% = \frac{105.5 - 57.2}{105.5} \times 100\% = \mathbf{45.8\%}$$

**Discussion** Note that the force required to push the block on the inclined surface reduces significantly by oiling the surface.



## 2-44

**Solution** The velocity profile of a fluid flowing through a circular pipe is given. The friction drag force exerted on the pipe by the fluid in the flow direction per unit length of the pipe is to be determined.

**Assumptions** The viscosity of the fluid is constant.

**Analysis** The wall shear stress is determined from its definition to be

$$\tau_w = -\mu \left. \frac{du}{dr} \right|_{r=R} = -\mu u_{\max} \frac{d}{dr} \left( 1 - \frac{r^n}{R^n} \right) \bigg|_{r=R} = -\mu u_{\max} \frac{-nr^{n-1}}{R^n} \bigg|_{r=R} = \frac{n\mu u_{\max}}{R}$$

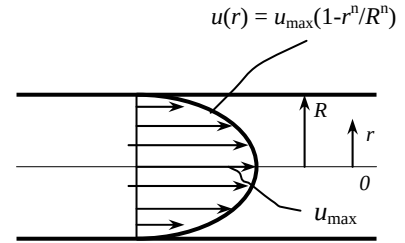
Note that the quantity  $du/dr$  is negative in pipe flow, and the negative sign is added to the  $\tau_w$  relation for pipes to make shear stress in the positive (flow) direction a positive quantity. (Or,  $du/dr = -du/dy$  since  $y = R - r$ ). Then the friction drag force exerted by the fluid on the inner surface of the pipe becomes

$$F = \tau_w A_w = \frac{n\mu u_{\max}}{R} (2\pi R) L = 2n\pi\mu u_{\max} L$$

Therefore, the drag force per unit length of the pipe is

$$\boxed{F/L = 2n\pi\mu u_{\max}}$$

**Discussion** Note that the drag force acting on the pipe in this case is independent of the pipe diameter.



## 2-45

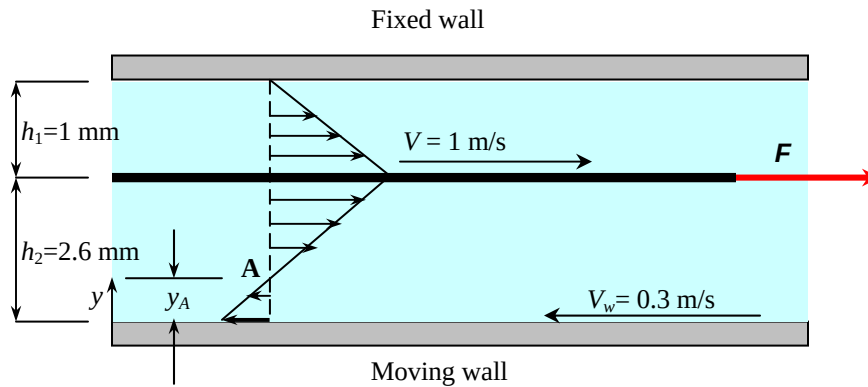
**Solution** A thin flat plate is pulled horizontally through an oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity. The location in oil where the velocity is zero and the force that needs to be applied on the plate are to be determined.

**Assumptions** 1 The thickness of the plate is negligible. 2 The velocity profile in each oil layer is linear.

**Properties** The absolute viscosity of oil is given to be  $\mu = 0.027 \text{ Pa}\cdot\text{s} = 0.027 \text{ N}\cdot\text{s}/\text{m}^2$ .

**Analysis** (a) The velocity profile in each oil layer relative to the fixed wall is as shown in the figure below. The point of zero velocity is indicated by point A, and its distance from the lower plate is determined from geometric considerations (the similarity of the two triangles in the lower oil layer) to be

$$\frac{2.6 - y_A}{y_A} = \frac{1}{0.3} \rightarrow y_A = \mathbf{0.60 \text{ mm}}$$



(b) The magnitudes of shear forces acting on the upper and lower surfaces of the plate are

$$F_{\text{shear, upper}} = \tau_{w, \text{upper}} A_s = \mu A_s \left| \frac{du}{dy} \right| = \mu A_s \frac{V - 0}{h_1} = (0.027 \text{ N}\cdot\text{s}/\text{m}^2)(0.2 \times 0.2 \text{ m}^2) \frac{1 \text{ m/s}}{1.0 \times 10^{-3} \text{ m}} = 1.08 \text{ N}$$

$$F_{\text{shear, lower}} = \tau_{w, \text{lower}} A_s = \mu A_s \left| \frac{du}{dy} \right| = \mu A_s \frac{V - V_w}{h_2} = (0.027 \text{ N}\cdot\text{s}/\text{m}^2)(0.2 \times 0.2 \text{ m}^2) \frac{[1 - (-0.3)] \text{ m/s}}{2.6 \times 10^{-3} \text{ m}} = 0.54 \text{ N}$$

Noting that both shear forces are in the opposite direction of motion of the plate, the force  $F$  is determined from a force balance on the plate to be

$$F = F_{\text{shear, upper}} + F_{\text{shear, lower}} = 1.08 + 0.54 = \mathbf{1.62 \text{ N}}$$

**Discussion** Note that wall shear is a friction force between a solid and a liquid, and it acts in the opposite direction of motion.

2-46

**Solution** A frustum shaped body is rotating at a constant angular speed in an oil container. The power required to maintain this motion and the reduction in the required power input when the oil temperature rises are to be determined.

**Assumptions** The thickness of the oil layer remains constant.

**Properties** The absolute viscosity of oil is given to be  $\mu = 0.1 \text{ Pa}\cdot\text{s} = 0.1 \text{ N}\cdot\text{s}/\text{m}^2$  at  $20^\circ\text{C}$  and  $0.0078 \text{ Pa}\cdot\text{s}$  at  $80^\circ\text{C}$ .

**Analysis** The velocity gradient anywhere in the oil of film thickness  $h$  is  $V/h$  where  $V = \omega r$  is the tangential velocity. Then the wall shear stress anywhere on the surface of the frustum at a distance  $r$  from the axis of rotation is

$$\tau_w = \mu \frac{du}{dr} = \mu \frac{V}{h} = \mu \frac{\omega r}{h}$$

The shear force acting on differential area  $dA$  on the surface, the torque it generates, and the shaft power associated with it are expressed as

$$dF = \tau_w dA = \mu \frac{\omega r}{h} dA \quad dT = r dF = \mu \frac{\omega r^2}{h} dA$$

$$T = \frac{\mu \omega}{h} \int_A r^2 dA \quad \dot{W}_{\text{sh}} = \omega T = \frac{\mu \omega^2}{h} \int_A r^2 dA$$

**Top surface:** For the top surface,  $dA = 2\pi r dr$ . Substituting and integrating,

$$\dot{W}_{\text{sh, top}} = \frac{\mu \omega^2}{h} \int_{r=0}^{D/2} r^2 (2\pi r) dr = \frac{2\pi \mu \omega^2}{h} \int_{r=0}^{D/2} r^3 dr = \frac{2\pi \mu \omega^2}{h} \frac{r^4}{4} \bigg|_{r=0}^{D/2} = \frac{\pi \mu \omega^2 D^4}{32h}$$

**Bottom surface:** A relation for the bottom surface is obtained by replacing  $D$  by  $d$ ,  $\dot{W}_{\text{sh, bottom}} = \frac{\pi \mu \omega^2 d^4}{32h}$

**Side surface:** The differential area for the side surface can be expressed as  $dA = 2\pi r dz$ . From geometric considerations, the variation of radius with axial distance is expressed as  $r = \frac{d}{2} + \frac{D-d}{2L} z$ .

Differentiating gives  $dr = \frac{D-d}{2L} dz$  or  $dz = \frac{2L}{D-d} dr$ . Therefore,  $dA = 2\pi r dz = \frac{4\pi L}{D-d} r dr$ . Substituting and integrating,

$$\dot{W}_{\text{sh, top}} = \frac{\mu \omega^2}{h} \int_{r=0}^{D/2} r^2 \frac{4\pi L}{D-d} r dr = \frac{4\pi \mu \omega^2 L}{h(D-d)} \int_{r=d/2}^{D/2} r^3 dr = \frac{4\pi \mu \omega^2 L}{h(D-d)} \frac{r^4}{4} \bigg|_{r=d/2}^{D/2} = \frac{\pi \mu \omega^2 L (D^2 - d^2)}{16h(D-d)}$$

Then the total power required becomes

$$\dot{W}_{\text{sh, total}} = \dot{W}_{\text{sh, top}} + \dot{W}_{\text{sh, bottom}} + \dot{W}_{\text{sh, side}} = \frac{\pi \mu \omega^2 D^4}{32h} \left[ 1 + (d/D)^4 + \frac{2L[1 - (d/D)^4]}{D-d} \right],$$

where  $d/D = 4/12 = 1/3$ . Substituting,

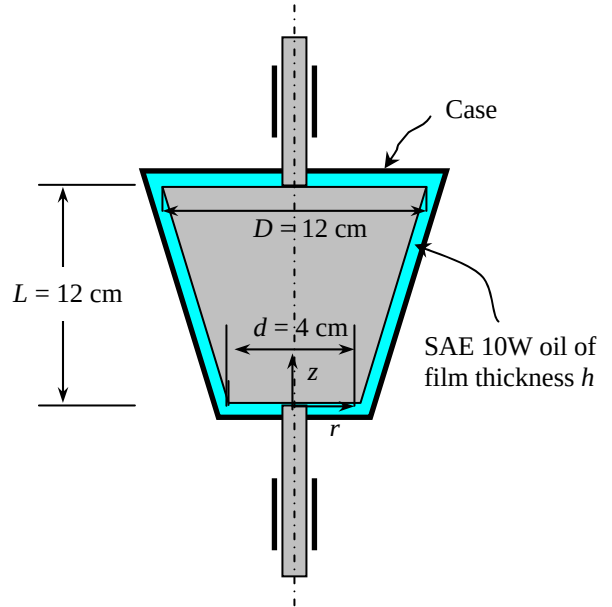
$$\dot{W}_{\text{sh, total}} = \frac{\pi (0.1 \text{ N}\cdot\text{s}/\text{m}^2) (200/\text{s})^2 (0.12 \text{ m})^4}{32(0.0012 \text{ m})} \left[ 1 + (1/3)^4 + \frac{2(0.12 \text{ m})[1 - (1/3)^4]}{(0.12 - 0.04) \text{ m}} \right] \left( \frac{1 \text{ W}}{1 \text{ N}\cdot\text{m}/\text{s}} \right) = 270 \text{ W}$$

Noting that power is proportional to viscosity, the power required at  $80^\circ\text{C}$  is

$$\dot{W}_{\text{sh, total, } 80^\circ\text{C}} = \frac{\mu_{80^\circ\text{C}}}{\mu_{20^\circ\text{C}}} \dot{W}_{\text{sh, total, } 20^\circ\text{C}} = \frac{0.0078 \text{ N}\cdot\text{s}/\text{m}^2}{0.1 \text{ N}\cdot\text{s}/\text{m}^2} (270 \text{ W}) = 21.1 \text{ W}$$

Therefore, the reduction in the requires power input at  $80^\circ\text{C}$  is Reduction =  $\dot{W}_{\text{sh, total, } 20^\circ\text{C}} - \dot{W}_{\text{sh, total, } 80^\circ\text{C}} = 270 - 21.1 = 249 \text{ W}$ , which is about 92%.

**Discussion** Note that the power required to overcome shear forces in a viscous fluid greatly depends on temperature.

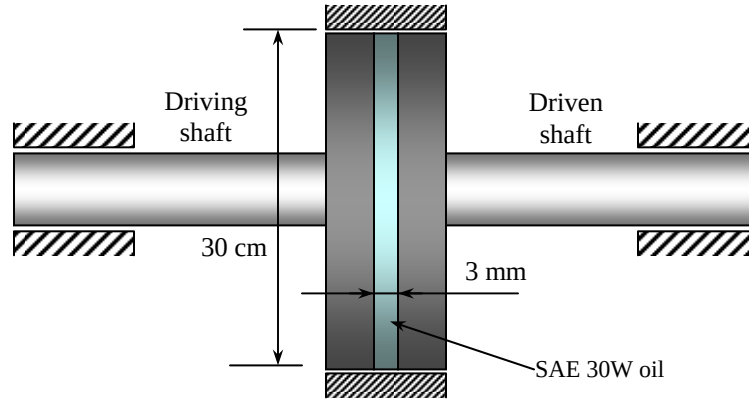


2-47

**Solution** A clutch system is used to transmit torque through an oil film between two identical disks. For specified rotational speeds, the transmitted torque is to be determined.

**Assumptions** 1 The thickness of the oil film is uniform. 2 The rotational speeds of the disks remain constant.

**Properties** The absolute viscosity of oil is given to be  $\mu = 0.38 \text{ N}\cdot\text{s}/\text{m}^2$ .



**Analysis** The disks are rotating in the same direction at different angular speeds of  $\omega_1$  and of  $\omega_2$ . Therefore, we can assume one of the disks to be stationary and the other to be rotating at an angular speed of  $\omega_1 - \omega_2$ . The velocity gradient anywhere in the oil of film thickness  $h$  is  $V/h$  where  $V = (\omega_1 - \omega_2)r$  is the tangential velocity. Then the wall shear stress anywhere on the surface of the faster disk at a distance  $r$  from the axis of rotation can be expressed as

$$\tau_w = \mu \frac{du}{dr} = \mu \frac{V}{h} = \mu \frac{(\omega_1 - \omega_2)r}{h}$$

Then the shear force acting on a differential area  $dA$  on the surface and the torque generation associated with it can be expressed as

$$dF = \tau_w dA = \mu \frac{(\omega_1 - \omega_2)r}{h} (2\pi r) dr$$

$$dT = r dF = \mu \frac{(\omega_1 - \omega_2)r^2}{h} (2\pi) dr = \frac{2\pi\mu(\omega_1 - \omega_2)}{h} r^3 dr$$

Integrating,

$$T = \frac{2\pi\mu(\omega_1 - \omega_2)}{h} \int_{r=0}^{D/2} r^3 dr = \frac{2\pi\mu(\omega_1 - \omega_2)}{h} \frac{r^4}{4} \Big|_{r=0}^{D/2} = \frac{\pi\mu(\omega_1 - \omega_2)D^4}{32h}$$

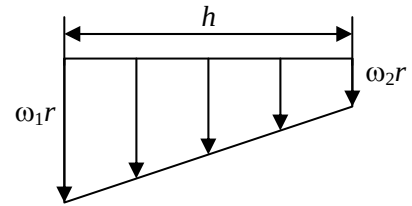
Noting that  $\omega = 2\pi \dot{n}$ , the relative angular speed is

$$\omega_1 - \omega_2 = 2\pi (\dot{n}_1 - \dot{n}_2) = (2\pi \text{ rad/rev}) [(1450 - 1398) \text{ rev/min}] \left( \frac{1 \text{ min}}{60 \text{ s}} \right) = 5.445 \text{ rad/s},$$

Substituting, the torque transmitted is determined to be

$$T = \frac{\pi(0.38 \text{ N}\cdot\text{s}/\text{m}^2)(5.445/\text{s})(0.30 \text{ m})^4}{32(0.003 \text{ m})} = \mathbf{0.55 \text{ N}\cdot\text{m}}$$

**Discussion** Note that the torque transmitted is proportional to the fourth power of disk diameter, and is inversely proportional to the thickness of the oil film.



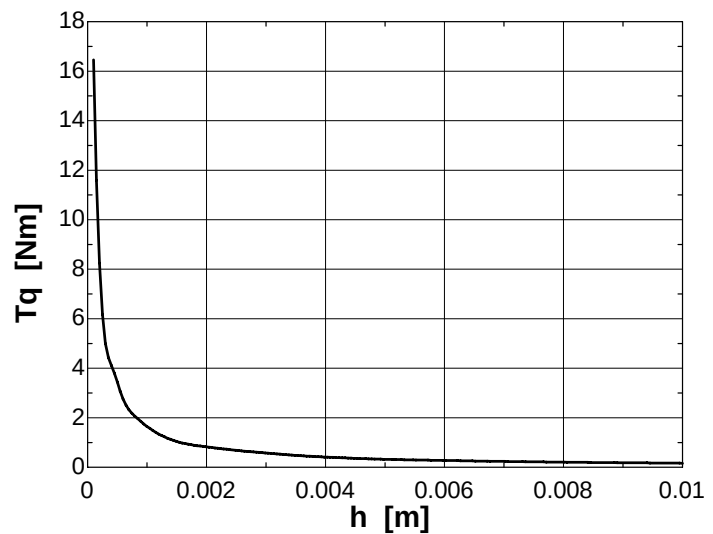


**Solution** We are to investigate the effect of oil film thickness on the transmitted torque.

**Analysis** The previous problem is reconsidered. Using EES software, the effect of oil film thickness on the torque transmitted is investigated. Film thickness varied from 0.1 mm to 10 mm, and the results are tabulated and plotted. The relation used is  $T = \frac{\pi\mu(\omega_1 - \omega_2)D^4}{32h}$ . The EES *Equations* window is printed below, followed by the tabulated and plotted results.

```
mu=0.38
n1=1450 "rpm"
w1=2*pi*n1/60 "rad/s"
n2=1398 "rpm"
w2=2*pi*n2/60 "rad/s"
D=0.3 "m"
Tq=pi*mu*(w1-w2)*(D^4)/(32*h)
```

| Film thickness<br><i>h</i> , mm | Torque transmitted<br><i>T</i> , Nm |
|---------------------------------|-------------------------------------|
| 0.1                             | 16.46                               |
| 0.2                             | 8.23                                |
| 0.4                             | 4.11                                |
| 0.6                             | 2.74                                |
| 0.8                             | 2.06                                |
| 1                               | 1.65                                |
| 2                               | 0.82                                |
| 4                               | 0.41                                |
| 6                               | 0.27                                |
| 8                               | 0.21                                |
| 10                              | 0.16                                |



**Conclusion** Torque transmitted is inversely proportional to oil film thickness, and the film thickness should be as small as possible to maximize the transmitted torque.

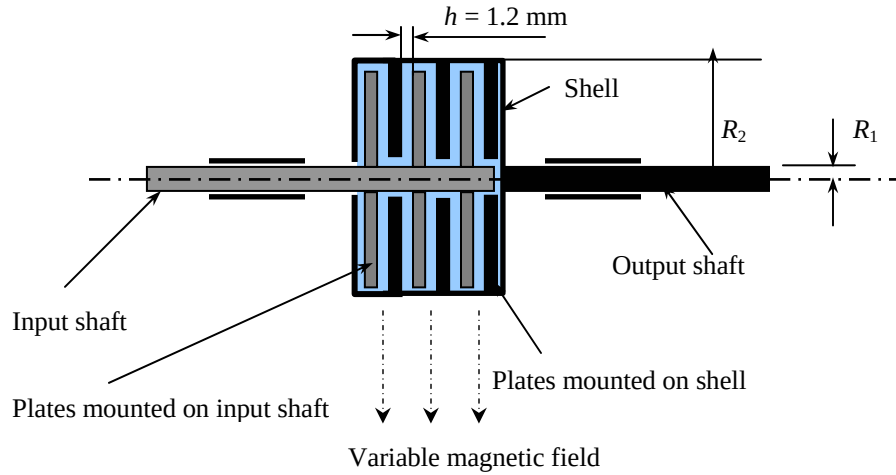
**Discussion** To obtain the solution in EES, we set up a parametric table, specify *h*, and let EES calculate *T* for each value of *h*.

**2-49**

**Solution** A multi-disk Electro-rheological “ER” clutch is considered. The ER fluid has a shear stress that is expressed as  $\tau = \tau_y + \mu(du/dy)$ . A relationship for the torque transmitted by the clutch is to be obtained, and the numerical value of the torque is to be calculated.

**Assumptions** 1 The thickness of the oil layer between the disks is constant. 2 The Bingham plastic model for shear stress expressed as  $\tau = \tau_y + \mu(du/dy)$  is valid.

**Properties** The constants in shear stress relation are given to be  $\mu = 0.1 \text{ Pa}\cdot\text{s}$  and  $\tau_y = 2.5 \text{ kPa}$ .



**Analysis** (a) The velocity gradient anywhere in the oil of film thickness  $h$  is  $V/h$  where  $V = \omega r$  is the tangential velocity relative to plates mounted on the shell. Then the wall shear stress anywhere on the surface of a plate mounted on the input shaft at a distance  $r$  from the axis of rotation is expressed as

$$\tau_w = \tau_y + \mu \frac{du}{dr} = \tau_y + \mu \frac{V}{h} = \tau_y + \mu \frac{\omega r}{h}$$

Then the shear force acting on a differential area  $dA$  on the surface of a disk and the torque generation associated with it are expressed as

$$dF = \tau_w dA = \left( \tau_y + \mu \frac{\omega r}{h} \right) (2\pi r) dr$$

$$dT = r dF = r \left( \tau_y + \mu \frac{\omega r}{h} \right) (2\pi r) dr = 2\pi \left( \tau_y r^2 + \mu \frac{\omega r^3}{h} \right) dr$$

Integrating,

$$T = 2\pi \int_{r=R_1}^{R_2} \left( \tau_y r^2 + \mu \frac{\omega r^3}{h} \right) dr = 2\pi \left[ \tau_y \frac{r^3}{3} + \frac{\mu \omega r^4}{4h} \right]_{r=R_1}^{R_2} = 2\pi \left[ \frac{\tau_y}{3} (R_2^3 - R_1^3) + \frac{\mu \omega}{4h} (R_2^4 - R_1^4) \right]$$

This is the torque transmitted by one surface of a plate mounted on the input shaft. Then the torque transmitted by both surfaces of  $N$  plates attached to input shaft in the clutch becomes

$$T = 4\pi N \left[ \frac{\tau_y}{3} (R_2^3 - R_1^3) + \frac{\mu \omega}{4h} (R_2^4 - R_1^4) \right]$$

(b) Noting that  $\omega = 2\pi \dot{m} = 2\pi(2400 \text{ rev/min}) = 15,080 \text{ rad/min} = 251.3 \text{ rad/s}$  and substituting,

$$T = (4\pi)(11) \left[ \frac{2500 \text{ N/m}^2}{3} [(0.20 \text{ m})^3 - (0.05 \text{ m})^3] + \frac{(0.1 \text{ N}\cdot\text{s/m}^2)(251.3/\text{s})}{4(0.0012 \text{ m})} [(0.20 \text{ m})^4 - (0.05 \text{ m})^4] \right] = \mathbf{2060 \text{ N}\cdot\text{m}}$$

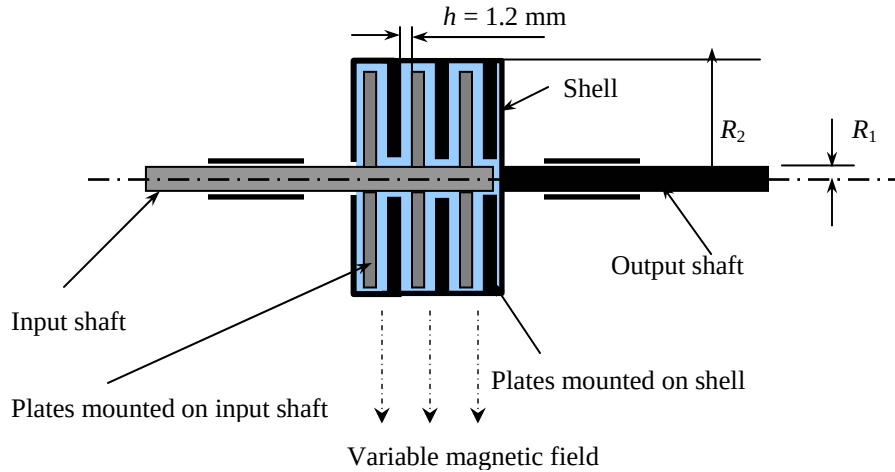
**Discussion** Can you think of some other potential applications for this kind of fluid?

## 2-50

**Solution** A multi-disk magnetorheological “MR” clutch is considered. The MR fluid has a shear stress that is expressed as  $\tau = \tau_y + K(du/dy)^m$ . A relationship for the torque transmitted by the clutch is to be obtained, and the numerical value of the torque is to be calculated.

**Assumptions** 1 The thickness of the oil layer between the disks is constant. 2 The Herschel-Bulkley model for shear stress expressed as  $\tau = \tau_y + K(du/dy)^m$  is valid.

**Properties** The constants in shear stress relation are given to be  $\tau_y = 900$  Pa,  $K = 58$  Pa·s<sup>m</sup>, and  $m = 0.82$ .



**Analysis** (a) The velocity gradient anywhere in the oil of film thickness  $h$  is  $V/h$  where  $V = \omega r$  is the tangential velocity relative to plates mounted on the shell. Then the wall shear stress anywhere on the surface of a plate mounted on the input shaft at a distance  $r$  from the axis of rotation is expressed as

$$\tau_w = \tau_y + K\left(\frac{du}{dr}\right)^m = \tau_y + K\left(\frac{V}{h}\right)^m = \tau_y + K\left(\frac{\omega r}{h}\right)^m$$

Then the shear force acting on a differential area  $dA$  on the surface of a disk and the torque generation associated with it are expressed as

$$dF = \tau_w dA = \left(\tau_y + K\left(\frac{\omega r}{h}\right)^m\right)(2\pi r)dr \quad \text{and} \quad dT = r dF = r\left(\tau_y + K\left(\frac{\omega r}{h}\right)^m\right)(2\pi r)dr = 2\pi\left(\tau_y r^2 + K\frac{\omega^m r^{m+2}}{h^m}\right)dr$$

Integrating,

$$T = 2\pi \int_{R_1}^{R_2} \left(\tau_y r^2 + K\frac{\omega^m r^{m+2}}{h^m}\right)dr = 2\pi \left[\tau_y \frac{r^3}{3} + \frac{K\omega^m r^{m+3}}{(m+3)h^m}\right]_{R_1}^{R_2} = 2\pi \left[\frac{\tau_y}{3}(R_2^3 - R_1^3) + \frac{K\omega^m}{(m+3)h^m}(R_2^{m+3} - R_1^{m+3})\right]$$

This is the torque transmitted by one surface of a plate mounted on the input shaft. Then the torque transmitted by both surfaces of  $N$  plates attached to input shaft in the clutch becomes

$$T = 4\pi N \left[\frac{\tau_y}{3}(R_2^3 - R_1^3) + \frac{K\omega^m}{(m+3)h^m}(R_2^{m+3} - R_1^{m+3})\right]$$

(b) Noting that  $\omega = 2\pi n = 2\pi(2400 \text{ rev/min}) = 15,080 \text{ rad/min} = 251.3 \text{ rad/s}$  and substituting,

$$\begin{aligned} T &= (4\pi)(11) \left[ \frac{900 \text{ N/m}^2}{3} [(0.20 \text{ m})^3 - (0.05 \text{ m})^3] + \frac{(58 \text{ N} \cdot \text{s}^{0.82}/\text{m}^2)(251.3 \text{ /s})^{0.82}}{(0.82+3)(0.0012 \text{ m})^{0.82}} [(0.20 \text{ m})^{3.82} - (0.05 \text{ m})^{3.82}] \right] \\ &= 103.4 \text{ N} \cdot \text{m} \approx \mathbf{103 \text{ kN} \cdot \text{m}} \end{aligned}$$

**Discussion** Can you think of some other potential applications for this kind of fluid?

## 2-51

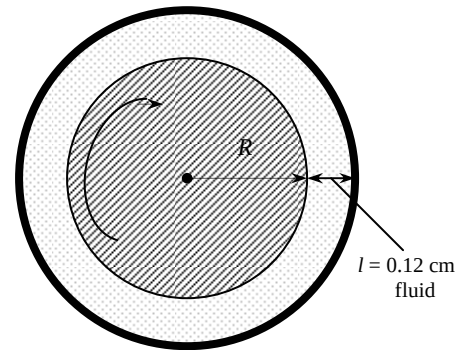
**Solution** The torque and the rpm of a double cylinder viscometer are given. The viscosity of the fluid is to be determined.

**Assumptions** 1 The inner cylinder is completely submerged in oil. 2 The viscous effects on the two ends of the inner cylinder are negligible. 3 The fluid is Newtonian.

**Analysis** Substituting the given values, the viscosity of the fluid is determined to be

$$\mu = \frac{T\ell}{4\pi^2 R^3 \dot{n}L} = \frac{(0.8 \text{ N} \cdot \text{m})(0.0012 \text{ m})}{4\pi^2 (0.075 \text{ m})^3 (200/60 \text{ s}^{-1})(0.75 \text{ m})} = 0.0231 \text{ N} \cdot \text{s}/\text{m}^2$$

**Discussion** This is the viscosity value at the temperature that existed during the experiment. Viscosity is a strong function of temperature, and the values can be significantly different at different temperatures.



## 2-52E

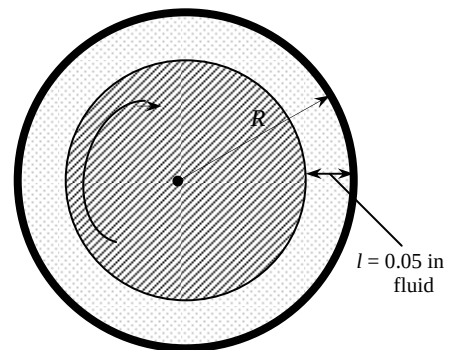
**Solution** The torque and the rpm of a double cylinder viscometer are given. The viscosity of the fluid is to be determined.

**Assumptions** 1 The inner cylinder is completely submerged in the fluid. 2 The viscous effects on the two ends of the inner cylinder are negligible. 3 The fluid is Newtonian.

**Analysis** Substituting the given values, the viscosity of the fluid is determined to be

$$\mu = \frac{T\ell}{4\pi^2 R^3 \dot{n}L} = \frac{(1.2 \text{ lbf} \cdot \text{ft})(0.05/12 \text{ ft})}{4\pi^2 (5.6/12 \text{ ft})^3 (250/60 \text{ s}^{-1})(3 \text{ ft})} = 9.97 \times 10^{-5} \text{ lbf} \cdot \text{s}/\text{ft}^2$$

**Discussion** This is the viscosity value at temperature that existed during the experiment. Viscosity is a strong function of temperature, and the values can be significantly different at different temperatures.





## 2-53

**Solution** The velocity profile for laminar one-dimensional flow through a circular pipe is given. A relation for friction drag force exerted on the pipe and its numerical value for water are to be determined.

**Assumptions** 1 The flow through the circular pipe is one-dimensional. 2 The fluid is Newtonian.

**Properties** The viscosity of water at 20°C is given to be 0.0010 kg/m·s.

**Analysis** The velocity profile is given by  $u(r) = u_{\max} \left( 1 - \frac{r^2}{R^2} \right)$

where  $R$  is the radius of the pipe,  $r$  is the radial distance from the center of the pipe, and  $u_{\max}$  is the maximum flow velocity, which occurs at the center,  $r = 0$ . The shear stress at the pipe surface is expressed as

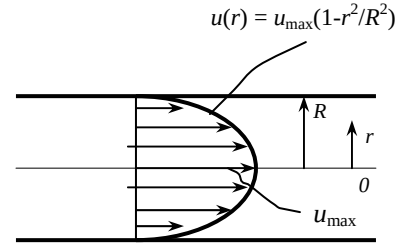
$$\tau_w = -\mu \left. \frac{du}{dr} \right|_{r=R} = -\mu u_{\max} \frac{d}{dr} \left( 1 - \frac{r^2}{R^2} \right) \bigg|_{r=R} = -\mu u_{\max} \frac{-2r}{R^2} \bigg|_{r=R} = \frac{2\mu u_{\max}}{R}$$

Note that the quantity  $du/dr$  is negative in pipe flow, and the negative sign is added to the  $\tau_w$  relation for pipes to make shear stress in the positive (flow) direction a positive quantity. (Or,  $du/dr = -du/dy$  since  $y = R - r$ ). Then the friction drag force exerted by the fluid on the inner surface of the pipe becomes

$$F_D = \tau_w A_s = \frac{2\mu u_{\max}}{R} (2\pi RL) = 4\pi \mu L u_{\max}$$

Substituting we get  $F_D = 4\pi \mu L u_{\max} = 4\pi (0.0010 \text{ kg/m} \cdot \text{s})(15 \text{ m})(3 \text{ m/s}) \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{0.565 \text{ N}}$

**Discussion** In the entrance region and during turbulent flow, the velocity gradient is greater near the wall, and thus the drag force in such cases will be greater.



## 2-54

**Solution** The velocity profile for laminar one-dimensional flow through a circular pipe is given. A relation for friction drag force exerted on the pipe and its numerical value for water are to be determined.

**Assumptions** 1 The flow through the circular pipe is one-dimensional. 2 The fluid is Newtonian.

**Properties** The viscosity of water at 20°C is given to be 0.0010 kg/m·s.

**Analysis** The velocity profile is given by  $u(r) = u_{\max} \left( 1 - \frac{r^2}{R^2} \right)$

where  $R$  is the radius of the pipe,  $r$  is the radial distance from the center of the pipe, and  $u_{\max}$  is the maximum flow velocity, which occurs at the center,  $r = 0$ . The shear stress at the pipe surface can be expressed as

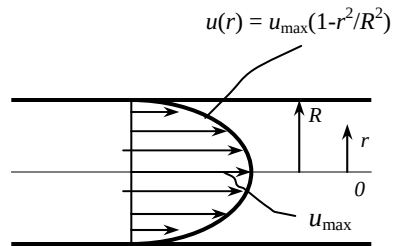
$$\tau_w = -\mu \left. \frac{du}{dr} \right|_{r=R} = -\mu u_{\max} \frac{d}{dr} \left( 1 - \frac{r^2}{R^2} \right) \bigg|_{r=R} = -\mu u_{\max} \frac{-2r}{R^2} \bigg|_{r=R} = \frac{2\mu u_{\max}}{R}$$

Note that the quantity  $du/dr$  is negative in pipe flow, and the negative sign is added to the  $\tau_w$  relation for pipes to make shear stress in the positive (flow) direction a positive quantity. (Or,  $du/dr = -du/dy$  since  $y = R - r$ ). Then the friction drag force exerted by the fluid on the inner surface of the pipe becomes

$$F_D = \tau_w A_s = \frac{2\mu u_{\max}}{R} (2\pi RL) = 4\pi \mu L u_{\max}$$

Substituting, we get  $F_D = 4\pi \mu L u_{\max} = 4\pi (0.0010 \text{ kg/m} \cdot \text{s})(15 \text{ m})(5 \text{ m/s}) \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{0.942 \text{ N}}$

**Discussion** In the entrance region and during turbulent flow, the velocity gradient is greater near the wall, and thus the drag force in such cases will be larger.



---

## Surface Tension and Capillary Effect

---

### 2-55C

**Solution** We are to define and discuss surface tension.

**Analysis** The **magnitude of the pulling force at the surface of a liquid per unit length** is called *surface tension*  $\sigma_s$ . It is caused by the attractive forces between the molecules. The surface tension is also surface energy (per unit area) since it represents the stretching work that needs to be done to increase the surface area of the liquid by a unit amount.

**Discussion** Surface tension is the cause of some very interesting phenomena such as capillary rise and insects that can walk on water.

---

### 2-56C

**Solution** We are to analyze the pressure difference between inside and outside of a soap bubble.

**Analysis** The **pressure inside a soap bubble is greater than the pressure outside**, as evidenced by the stretch of the soap film.

**Discussion** You can make an analogy between the soap film and the skin of a balloon.

---

### 2-57C

**Solution** We are to define and discuss the capillary effect.

**Analysis** The *capillary effect* is the **rise or fall of a liquid in a small-diameter tube inserted into the liquid**. It is caused by the net effect of the *cohesive forces* (the forces between like molecules, like water) and *adhesive forces* (the forces between unlike molecules, like water and glass). **The capillary effect is proportional to the cosine of the contact angle**, which is the angle that the tangent to the liquid surface makes with the solid surface at the point of contact.

**Discussion** The contact angle determines whether the *meniscus* at the top of the column is concave or convex.

---

### 2-58C

**Solution** We are to determine whether the level of liquid in a tube will rise or fall due to the capillary effect.

**Analysis** The **liquid level in the tube will drop** since the contact angle is greater than  $90^\circ$ , and  $\cos(110^\circ) < 0$ .

**Discussion** This liquid must be a non-wetting liquid when in contact with the tube material. Mercury is an example of a non-wetting liquid with a contact angle (with glass) that is greater than  $90^\circ$ .

---

### 2-59C

**Solution** We are to compare the capillary rise in small and large diameter tubes.

**Analysis** The capillary rise is inversely proportional to the diameter of the tube, and thus **capillary rise is greater in the smaller-diameter tube**.

**Discussion** Note however, that if the tube diameter is large enough, there is no capillary rise (or fall) at all. Rather, the upward (or downward) rise of the liquid occurs only near the tube walls; the elevation of the middle portion of the liquid in the tube does not change for large diameter tubes.

---

## 2-60E

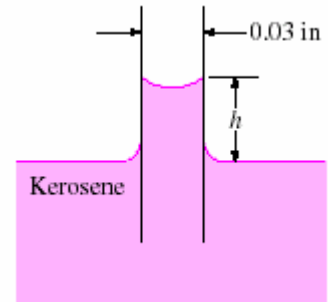
**Solution** A slender glass tube is inserted into kerosene. The capillary rise of kerosene in the tube is to be determined.

**Assumptions** 1 There are no impurities in the kerosene, and no contamination on the surfaces of the glass tube. 2 The kerosene is open to the atmospheric air.

**Properties** The surface tension of kerosene-glass at 68°F (20°C) is  $\sigma_s = 0.028 \times 0.06852 = 0.00192$  lbf/ft. The density of kerosene at 68°F is  $\rho = 51.2$  lbm/ft<sup>3</sup>. The contact angle of kerosene with the glass surface is given to be 26°.

**Analysis** Substituting the numerical values, the capillary rise is determined to be

$$h = \frac{2\sigma_s \cos \phi}{\rho g R} = \frac{2(0.00192 \text{ lbf/ft})(\cos 26^\circ)}{(51.2 \text{ lbm/ft}^3)(32.2 \text{ ft/s}^2)(0.015/12 \text{ ft})} \left( \frac{32.2 \text{ lbm} \cdot \text{ft/s}^2}{1 \text{ lbf}} \right) \\ = 0.0539 \text{ ft} = \mathbf{0.650 \text{ in}}$$



**Discussion** The capillary rise in this case more than half of an inch, and thus it is clearly noticeable.

## 2-61

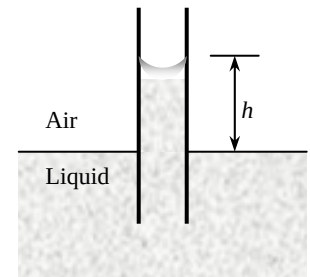
**Solution** A glass tube is inserted into a liquid, and the capillary rise is measured. The surface tension of the liquid is to be determined.

**Assumptions** 1 There are no impurities in the liquid, and no contamination on the surfaces of the glass tube. 2 The liquid is open to the atmospheric air.

**Properties** The density of the liquid is given to be 960 kg/m<sup>3</sup>. The contact angle is given to be 15°.

**Analysis** Substituting the numerical values, the surface tension is determined from the capillary rise relation to be

$$\sigma_s = \frac{\rho g R h}{2 \cos \phi} = \frac{(960 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.0019/2 \text{ m})(0.005 \text{ m})}{2(\cos 15^\circ)} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{0.0232 \text{ N/m}}$$



**Discussion** Since surface tension depends on temperature, the value determined is valid at the liquid's temperature.

## 2-62

**Solution** The diameter of a soap bubble is given. The gage pressure inside the bubble is to be determined.

**Assumptions** The soap bubble is in atmospheric air.

**Properties** The surface tension of soap water at 20°C is  $\sigma_s = 0.025$  N/m.

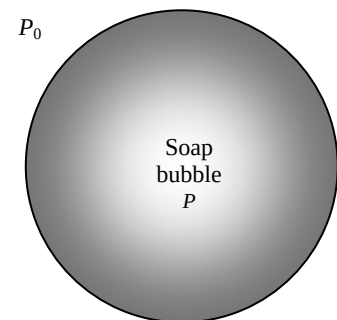
**Analysis** The pressure difference between the inside and the outside of a bubble is given by

$$\Delta P_{\text{bubble}} = P_i - P_0 = \frac{4\sigma_s}{R}$$

In the open atmosphere  $P_0 = P_{\text{atm}}$ , and thus  $\Delta P_{\text{bubble}}$  is equivalent to the gage pressure. Substituting,

$$P_{i,\text{gage}} = \Delta P_{\text{bubble}} = \frac{4(0.025 \text{ N/m})}{0.002/2 \text{ m}} = 100 \text{ N/m}^2 = 100 \text{ Pa}$$

$$P_{i,\text{gage}} = \Delta P_{\text{bubble}} = \frac{4(0.025 \text{ N/m})}{0.05/2 \text{ m}} = 4 \text{ N/m}^2 = 4 \text{ Pa}$$



**Discussion** Note that the gage pressure in a soap bubble is inversely proportional to the radius. Therefore, the excess pressure is larger in smaller bubbles.

### 2-63

**Solution** Nutrients dissolved in water are carried to upper parts of plants. The height to which the water solution rises in a tree as a result of the capillary effect is to be determined.

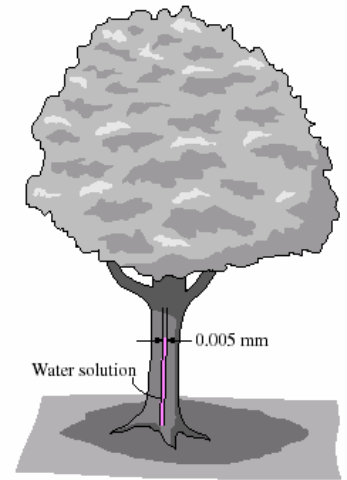
**Assumptions** 1 The solution can be treated as water with a contact angle of  $15^\circ$ . 2 The diameter of the tube is constant. 3 The temperature of the water solution is  $20^\circ\text{C}$ .

**Properties** The surface tension of water at  $20^\circ\text{C}$  is  $\sigma_s = 0.073 \text{ N/m}$ . The density of water solution can be taken to be  $1000 \text{ kg/m}^3$ . The contact angle is given to be  $15^\circ$ .

**Analysis** Substituting the numerical values, the capillary rise is determined to be

$$h = \frac{2\sigma_s \cos \phi}{\rho g R} = \frac{2(0.073 \text{ N/m})(\cos 15^\circ)}{(1000 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(2.5 \times 10^{-6} \text{ m})} \left( \frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}} \right) = 5.75 \text{ m}$$

**Discussion** Other effects such as the chemical potential difference also cause the fluid to rise in trees.



### 2-64

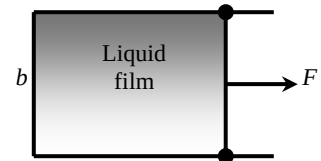
**Solution** The force acting on the movable wire of a liquid film suspended on a U-shaped wire frame is measured. The surface tension of the liquid in the air is to be determined.

**Assumptions** 1 There are no impurities in the liquid, and no contamination on the surfaces of the wire frame. 2 The liquid is open to the atmospheric air.

**Analysis** Substituting the numerical values, the surface tension is determined from the surface tension force relation to be

$$\sigma_s = \frac{F}{2b} = \frac{0.012 \text{ N}}{2(0.08 \text{ m})} = 0.075 \text{ N/m}$$

**Discussion** The surface tension depends on temperature. Therefore, the value determined is valid at the temperature of the liquid.



## 2-65

**Solution** A steel ball floats on water due to the surface tension effect. The maximum diameter of the ball is to be determined, and the calculations are to be repeated for aluminum.

**Assumptions** 1 The water is pure, and its temperature is constant. 2 The ball is dropped on water slowly so that the inertial effects are negligible. 3 The contact angle is taken to be  $0^\circ$  for maximum diameter.

**Properties** The surface tension of water at  $20^\circ\text{C}$  is  $\sigma_s = 0.073 \text{ N/m}$ . The contact angle is taken to be  $0^\circ$ . The densities of steel and aluminum are given to be  $\rho_{\text{steel}} = 7800 \text{ kg/m}^3$  and  $\rho_{\text{Al}} = 2700 \text{ kg/m}^3$ .

**Analysis** The surface tension force and the weight of the ball can be expressed as

$$F_s = \pi D \sigma_s \quad \text{and} \quad W = mg = \rho g V = \rho g \pi D^3 / 6$$

When the ball floats, the net force acting on the ball in the vertical direction is zero. Therefore, setting  $F_s = W$  and solving

for diameter  $D$  gives  $D = \sqrt{\frac{6\sigma_s}{\rho g}}$ . Substituting the known quantities, the maximum diameters for the steel and aluminum balls become

$$D_{\text{steel}} = \sqrt{\frac{6\sigma_s}{\rho g}} = \sqrt{\frac{6(0.073 \text{ N/m})}{(7800 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left( \frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}} \right)} = 2.4 \times 10^{-3} \text{ m} = \mathbf{2.4 \text{ mm}}$$

$$D_{\text{Al}} = \sqrt{\frac{6\sigma_s}{\rho g}} = \sqrt{\frac{6(0.073 \text{ N/m})}{(2700 \text{ kg/m}^3)(9.81 \text{ m/s}^2)} \left( \frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}} \right)} = 4.1 \times 10^{-3} \text{ m} = \mathbf{4.1 \text{ mm}}$$

**Discussion** Note that the ball diameter is inversely proportional to the square root of density, and thus for a given material, the smaller balls are more likely to float.

